

MATH 4650 – Numerical Analysis I

CSCI 4650 – Numerical Analysis I

Summer 2021

Sections MATH 4650-E01 and CSCI 4650-001

Department of Mathematical and Statistical Sciences, University of Colorado Denver

Instructor information

Instructor: Julien Langou, email: julien.langou@ucdenver.edu or through Canvas.

Office hours: By appointment. I will try to be as flexible as possible while trying to maximize the attendance at my office hour.

Course catalog description and requisites

MATH 4650 - Numerical Analysis I - 3 Credits

Methods and analysis of techniques used to resolve continuous mathematical problems on the computer. Solution of linear and nonlinear equations, interpolation and integration.

Prereq: MATH 3191 or MATH 3195 with a C- or higher.

CSCI 4650 - Numerical Analysis I - 3 Credits

Methods and analysis of techniques used to resolve continuous mathematical problems on the computer. Solution of linear and nonlinear equations, interpolation and integration.

Prereq: MATH 2411, MATH 3191 or MATH 3195, and programming experience.

Course Goals

Completion of this course will provide you with

1. an understanding of the basic theory of solving mathematical problems with computers while being cognizant of floating-point arithmetic including issues of overflow and underflow.
2. knowledge of the different issues surrounding errors in using numerical methods including machine epsilon, error analysis, convergence, rounding error, truncation error, and norms.
3. an appreciation of the difficulties involved in finding reliable solutions as well as be able to apply various methods for estimating errors in solutions in order to judge how reliable those solutions are.
4. an awareness of conditioning of problems and stability of algorithms and the distinguish between the two.

Programming Language

We will be using Python through Google Colab Jupyter Notebook (<https://colab.research.google.com/>).

Course materials and procedures

Textbook. *Numerical Analysis, 3rd edition* by Tim Sauer, ISBN-13: 978-0134696454, ISBN-10: 9780134696454, published by Pearson, October 30th, 2017. The link to the textbook is <https://www.pearson.com/us/higher-education/program/Sauer-Numerical-Analysis-3rd-Edition/PGM1735484.html>.

The link for purchase options is <https://www.pearson.com/store/p/numerical-analysis-/P100000809731>. You can purchase a 180 days subscription for \$39.99 + tax or a lifetime access \$59.99 + tax.

More procedures. Assignments and additional course materials will be posted on Canvas. Announcements, including any revisions to this syllabus, will be announced and posted on Canvas.

Evaluation

During the semester, grades will be posted on Canvas. Students should regularly check their recorded grades, and immediately bring any discrepancies or disputes to the attention of the instructor. Note that the grade calculation capabilities of Canvas are limited and may not be accurate. Use the grading scheme as described in this syllabus to compute your course grade.

You are encouraged to work together in groups outside of class, as well as consult other resources. However, your solutions must be your own. Any submitted solutions that I feel have been mostly copied from other sources, including classmates, textbooks, or the web, will receive no credit. See also the later discussion on academic honesty.

Homework ($12.5\% \times 6 = 75\%$). There will be 6 homework during the semester. Each homework is worth 12.5% of the final grade. For each homework, you need to turn in a “Google Colab Jupyter Notebook shared link”, a “.IPYNB” file, and a PDF with handwritten solutions.

Project ($12.5\% \times 2 = 25\%$). There will be 2 projects during the semester. Each project is worth 12.5% of the final grade. For each reality check, you need to turn in a “Google Colab Jupyter Notebook shared link”, a “.IPYNB” file, and a “.PDF” with handwritten solutions.

Final course grade scale. Final course letter grades will be assigned according to a student’s total course score (calculated as described above). Letter grades for specific scores are given in the following table.

$\geq 93\%$	A	$\geq 87\%$	B+	$\geq 77\%$	C+	$\geq 60\%$	D	$< 60\%$	F
$\geq 90\%$	A-	$\geq 83\%$	B	$\geq 70\%$	C				
		$\geq 80\%$	B-						

Due Dates.

Homework #1	Saturday June 12th 2021	at 11:59pm
Homework #2	Saturday June 19th 2021	at 11:59pm
Homework #3	Saturday June 26th 2021	at 11:59pm
Project #1	Saturday July 3rd 2021	at 11:59pm
Homework #4	Saturday July 10th 2021	at 11:59pm
Homework #5	Saturday July 17th 2021	at 11:59pm
Homework #6	Saturday July 24th 2021	at 11:59pm
Project #2	Saturday July 31st 2021	at 11:59pm

Chapter 0 Fundamentals		
0.1 Evaluating a Polynomial	week 1	lecture 1
0.2 Binary Numbers	week 1	lecture 2
0.2.1 - Decimal to binary	week 1	lecture 2
0.2.2 - Binary to decimal	week 1	lecture 2
0.3 Floating Point Representation of Real Numbers	week 1	lecture 3
0.3.1 - Floating point formats	week 1	lecture 3
0.3.2 - Machine representation	week 1	lecture 3
0.3.3 - Addition of floating point numbers	week 1	lecture 3
0.4 Loss of Significance	week 1	lecture 4
0.5 Review of Calculus	week 1	lecture 5
Chapter 1 Solving Equations		
1.1 The Bisection Method	week 2	lecture 6
1.1.1 - Bracketing a root	week 2	lecture 6
1.1.2 - How accurate and how fast?	week 2	lecture 6
1.2 Fixed-Point Iteration	week 2	lecture 9
1.2.1 - Fixed points of a function	week 2	lecture 9
1.2.2 - Geometry of Fixed-Point Iteration	week 2	lecture 9
1.2.3 - Linear convergence of Fixed-Point Iteration	week 2	lecture 9
1.2.4 - Stopping criteria	week 2	lecture 9
1.3 Limits of Accuracy	week 2	lecture 7
1.3.1 - Forward and backward error	week 2	lecture 7
1.3.2 - The Wilkinson polynomial	week 2	lecture 7
1.3.3 - Sensitivity of root-finding	week 2	lecture 7
1.4 Newton's Method	week 2	lecture 8
1.4.1 - Quadratic convergence of Newton's Method	week 2	lecture 8
1.4.2 - Linear convergence of Newton's Method	week 2	lecture 8
1.5 Root-Finding without Derivatives	week 2	lecture 10
1.5.1 - Secant Method and variants	week 2	lecture 10
1.5.2 - Brent's Method	week 2	lecture 10
Reality Check 1 Kinematics of the Stewart platform	week 4	
Chapter 2 Systems of Equations		
2.1 Gaussian Elimination	week 3	lecture 11
2.1.1 - Naïve Gaussian elimination	week 3	lecture 11
2.1.2 - Operation counts	week 3	lecture 11
2.2 The LU Factorization	week 3	lecture 12
2.2.1 - Matrix form of Gaussian elimination	week 3	lecture 12
2.2.2 - Back substitution with the LU factorization	week 3	lecture 12
2.2.3 - Complexity of the LU factorization	week 3	lecture 12
2.3 Sources of Error	week 3	lecture 13
2.3.1 - Error magnification and condition number	week 3	lecture 13
2.3.2 - Swamping	week 3	lecture 13
2.4 The PA=LU Factorization	week 3	lecture 14
2.4.1 - Partial pivoting	week 3	lecture 14
2.4.2 - Permutation matrices	week 3	lecture 14
2.4.3 - PA=LU factorization	week 3	lecture 14
Reality Check 2 The Euler-Bernoulli Beam	week 4	
2.5 Iterative Methods		
2.5.1 - Jacobi Method		
2.5.2 - Gauss-Seidel Method and SOR		
2.5.3 - Convergence of iterative methods		
2.5.4 - Sparse matrix computations		
2.6 Methods for symmetric positive definite matrices		
2.6.1 - Symmetric positive definite matrices		
2.6.2 - Cholesky factorization		
2.6.3 - Conjugate Gradient Method		
2.6.4 - Preconditioning		
2.7 Nonlinear Systems of Equations	week 3	lecture 15
2.7.1 - Multivariate Newton's Method	week 3	lecture 15
2.7.2 - Broyden's Method	week 3	lecture 15
Chapter 3 Interpolation		
3.1 Data and Interpolating Functions	week 5	lecture 16
3.1.1 - Lagrange interpolation	week 5	lecture 16
3.1.2 - Newton's divided differences	week 5	lecture 16
3.1.3 - How many degree d polynomials pass through n points?	week 5	lecture 16
3.1.4 - Code for interpolation	week 5	lecture 16
3.1.5 - Representing functions by approximating polynomials	week 5	lecture 16
3.2 Interpolation Error	week 5	lecture 17
3.2.1 - Interpolation error formula	week 5	lecture 17
3.2.2 - Proof of Newton form and error formula	week 5	lecture 17
3.2.3 - Runge phenomenon	week 5	lecture 17
3.3 Chebyshev interpolation	week 5	lecture 18
3.3.1 - Chebyshev's theorem	week 5	lecture 18
3.3.2 - Chebyshev polynomials	week 5	lecture 18
3.3.3 - Change of interval	week 5	lecture 18
2.4 Cubic Splines		
2.4.1 - Properties of splines		
2.4.2 - Endpoint conditions		
2.5 Bézier Curves		
Reality Check 3 Fonts from Bézier curves		
Chapter 4 Least Squares		
4.1 Least Squares and the Normal Equations	week 6	lecture 19
4.1.1 - Inconsistent systems of equations	week 6	lecture 19
4.1.2 - Fitting models to data	week 6	lecture 19
4.1.3 - Conditioning of least squares	week 6	lecture 19
4.2 A Survey of Models	week 6	lecture 20
4.2.1 - Periodic data	week 6	lecture 20
4.2.2 - Data linearization	week 6	lecture 20
4.3 QR Factorization	week 6	lecture 21
4.3.1 - Gram-Schmidt orthogonalization and least squares	week 6	lecture 21
4.3.2 - Modified Gram-Schmidt orthogonalization		
4.3.2 - Householder reflections		
4.4 Generalized Minimum Residual (GMRES) Method		
4.4.1 - Krylov methods		
4.4.2 - Preconditioned GMRES		
4.5 Nonlinear Least Squares	week 6	lecture 22
4.5.1 - Gauss-Newton Method	week 6	lecture 22
4.5.2 - Models with nonlinear parameters	week 6	lecture 22
4.5.3 - The Levenberg-Marquardt Method	week 6	lecture 22
Reality Check 4 GPS, Conditioning, and Nonlinear Least Squares	week 8	
Chapter 5 Numerical Differentiation and Integration	week 7	
5.1 Numerical Differentiation	week 7	lecture 23
5.1.1 - Finite difference formulas	week 7	lecture 23
5.1.2 - Rounding error	week 7	lecture 23
5.1.3 - Extrapolation	week 7	lecture 23
5.1.4 - Symbolic differentiation and integration	week 7	lecture 23
5.2 Newton-Cotes Formulas for Numerical Integration	week 7	lecture 24
5.2.1 - Trapezoid Rule	week 7	lecture 24
5.2.2 - Simpson's Rule	week 7	lecture 24
5.2.3 - Composite Newton-Cotes formulas	week 7	lecture 24
5.2.4 - Open Newton-Cotes Methods	week 7	lecture 24
5.3 Romberg Integration	week 7	lecture 25
5.4 Adaptive Quadrature	week 7	lecture 26
5.5 Gaussian Quadrature	week 7	lecture 27
Reality Check 5 Motion Control in Computer-Aided Modeling	week 8	

University, college, and department policies

Academic Calendar

For university deadlines and procedures (such as the last day to withdraw from a course), please see the Academic Calendar. <https://www.ucdenver.edu/student/registration-planning/academic-calendars>

Academic Support

Instructor office hours or other appointments are the best way to get additional help. I'm happy to help with questions not answered during lecture, additional explanation, or homework assistance.

The Math Education Resource Center (MERC Lab) located in North Classroom Room 4013 provides math tutoring. <https://clas.ucdenver.edu/mathematical-and-statistical-sciences/merc-lab>

The Learning Resources Center (LRC) provides individual and group tutoring, Supplemental Instruction (SI), study skills workshops, and ESL support. <https://www.ucdenver.edu/learning-resources-center>

The College of Liberal Arts and Sciences has a summary of campus academic support and school/college advising offices. <https://clas.ucdenver.edu/faculty-staff/content/clas-academic-policies-deadlines>

Disability Accommodation and Access

The University of Colorado Denver is committed to ensuring the full participation of all students in its programs, including students with disabilities. If you have a disability or think you have a disability and need accommodations to succeed in this course, I encourage you to contact Disability Resources and Services (DRS) and/or speak with me as soon as you can. DRS is located in Student Commons Building Suite 2116, and can be reached at disabilityresources@ucdenver.edu and online at <https://www.ucdenver.edu/student-services/resources/disability-resources-services>. I am committed to providing equal access as required by federal law, and I am interested in developing strategies for your success in this course.

Nondiscrimination and Sexual Misconduct

The University of Colorado Denver is committed to maintaining a positive learning, working and living environment. University policy and Title IX prohibit discrimination on the basis of race, color, national origin, sex, age, disability, pregnancy, creed, religion, sexual orientation, veteran status, gender identity, gender expression, political philosophy or political affiliation in admission and access to, and treatment and employment in, its educational programs and activities. University policy prohibits sexual misconduct, including harassment, domestic and dating violence, sexual assault, stalking, or related retaliation. If you have experienced any sort of sexual misconduct or discrimination, please visit the Office of Equity web site at <https://www.ucdenver.edu/offices/equity> to understand the resources available to you or contact the Office of Equity/Title IX Coordinator at equity@ucdenver.edu.

Please note that I am a **Responsible Employee**, which means that if I witness or receive information regarding possible prohibited protected characteristic discrimination or harassment, any form of sexual misconduct, and/or related retaliation, I am required to promptly report the information to the Office of Equity or their designee.

Religious Holiday Accommodation

Faculty in the University of Colorado system provide reasonable accommodations to students who must be absent from classes because of religious holidays. If you will miss class or graded assignments in order to observe religious holidays, you must contact me with all course conflicts by the end of the first week of classes.

Student Code of Conduct

As members of the University community, students are expected to uphold university standards, which include abiding by state civil and criminal laws and all University policies and standards of conduct. These standards are outlined in the student code of conduct, which can be found at <https://www.ucdenver.edu/student/health-wellness/student-conduct>

Academic Honesty

Students are expected to know, understand, and comply with the ethical standards of the university. A university's reputation is built on a standing tradition of excellence and scholastic integrity. As members of the University of Colorado Denver academic community, faculty and students accept the responsibility to maintain the highest standards of intellectual honesty and ethical conduct.

Academic dishonesty is defined as a student's use of unauthorized assistance with intent to deceive an instructor or other such person who may be assigned to evaluate the student's work in meeting course and degree requirements.

This course assumes your knowledge of the policies and [definitions](#). University policies allow the instructor to decide how to respond to an ethics violation, whether by lowering the assignment grade, lowering the course grade, and/or filing charges against the student with the Academic Ethics Committee. For more information regarding the CLAS Academic Ethics Committee policies and procedures, please refer to <https://clas.ucdenver.edu/faculty-staff/policies-procedures/handling-academic-dishonesty>. Violating the academic honor code can lead to expulsion from the University.

Examples of academic dishonesty include, but are not limited to, the following:

Plagiarism. Plagiarism is the use of another person's distinctive words or ideas without acknowledgment. Examples include:

1. Word-for-word copying of another person's ideas or words;
2. The mosaic (the interspersing of one's own words here and there while, in essence, copying another's work);
3. The paraphrase (the rewriting of another's work, yet still using their fundamental idea or theory);
4. Fabrication of references (inventing or counterfeiting sources);
5. Submission of another's work as one's own;
6. Neglecting quotation marks on material that is otherwise acknowledged.

Acknowledgment is not necessary when the material used is common knowledge.

Cheating. Cheating involves the possession, communication, or use of information, materials, notes, study aids or other devices not authorized by the instructor in an academic exercise, or communication with another person during such an exercise. Examples include:

1. Copying from another's paper or receiving unauthorized assistance from another during an academic exercise or in the submission of academic material;
2. Using a calculator when its use has been disallowed;
3. Collaborating with another student or students during an academic exercise without the consent of the instructor.

Fabrication and Falsification. Fabrication involves inventing or counterfeiting information, i.e., creating results not obtained in a study or laboratory experiment. Falsification, on the other hand, involves the deliberate alteration of results to suit one's needs in an experiment or other academic exercise.

Multiple Submissions. This is the submission of academic work for which academic credit has already been earned, when such submission is made without instructor authorization.

Misuse of Academic Materials. The misuse of academic materials includes, but is not limited to, the following:

1. Stealing or destroying library or reference materials or computer programs;
2. Stealing or destroying another student's notes or materials, or having such materials in one's possession without the owner's permission;

3. Receiving assistance in locating or using sources of information in an assignment when such assistance has been forbidden by the instructor;
4. Illegitimate possession, disposition, or use of examinations or answer keys to examinations;
5. Unauthorized alteration, forgery, or falsification;
6. Unauthorized sale or purchase of examinations, papers, or assignments.

Complicity in Academic Dishonesty. Complicity involves knowingly contributing to another's acts of academic dishonesty. Examples include:

1. Knowingly aiding another in any act of academic dishonesty;
2. Allowing another to copy from one's paper for an assignment or exam;
3. Distributing test questions or information about the materials to be tested before the scheduled exercise;
4. Taking an exam or test for someone else;
5. Signing another's name on attendance roster or on an academic exercise.

Incomplete Policy

When a student has special circumstances that make it impossible to complete course assignments, faculty members may choose to award an incomplete grade. All incomplete courses are assigned a grade of Incomplete (I). Incomplete grades are not awarded for poor academic performance or as a way of extending assignment deadlines. Faculty are not required to award an Incomplete.

To be eligible for an Incomplete grade, students **MUST**:

- Have participated in the class for a significant proportion of the term.
- Have successfully completed a significant proportion of the course assignments.
- Have special circumstances (verification may be required) that preclude the student from attending class and/or completing graded assignments.
- Make arrangements to complete missing assignments with the original instructor by a mutually agreed upon date but within one calendar year. Both the instructor and student should complete and sign a [CLAS Course Completion Agreement](#), and give a copy to the department.

Incompletes cannot:

- require a student to repeat the entire course,
- repeat or replace existing grades,
- allow the student an indeterminate period of time to complete a course, or
- allow the student to repeat the course with a different instructor.

Student Grievances

Students who have concerns about the course or instructor should first contact the instructor to discuss the issue. If the issue is not resolved, the student should next contact the Associate Chair of the Department of Mathematical and Statistical Sciences (currently Stephen Hartke, stephen.hartke@ucdenver.edu). If not satisfied, the student should then appeal to the appropriate Associate Dean of the student's home school or college (for CLAS, this is the Associate Dean for Student Success). No step in this process should be skipped.

STUDENT SUPPORT

CARE Team is there for you
Call 303-352-3579 if you
or a classmate needs extra help

Submit a concern at
<http://www.ucdenver.edu/care>

Call 911 in case of emergency
Auraria Campus Police: 303-556-5000

CAREER COUNSELING at LYNXCONNECT

ucdenver.edu/careercenter - Tivoli 339
303-315-7315 - CareerCenter@ucdenver.edu

COUNSELING CENTER

ucdenver.edu/counselingcenter - Tivoli 454 (4th floor)
303-315-7270 (Emergency After-Hours: 303-615-9911)

DISABILITY RESOURCES & SERVICES

ucdenver.edu/student-services/resources/disability-resources-services
Student Commons 2116
303-315-3510 - disabilityresources@ucdenver.edu

OFFICE OF EQUITY

equity.ucdenver.edu - Lawrence Street Center 12th floor
303-315-2567 – equity@ucdenver.edu

PHOENIX CENTER AT AURARIA

24/7 Free and Confidential Helpline: 303-556-2255
Info on interpersonal violence, referrals, options, & next steps
www.thepca.org - Tivoli 227 - 303-315-7250 - info@thepca.org

FREE TUTORING

Contact these services for academic
assistance throughout the semester

LEARNING RESOURCES CENTER

ucdenver.edu/lrc - Student Commons 2105
303-315-3531 - tutorialservices@ucdenver.edu

MATH EDUCATION RESOURCE CENTER

MERC Lab North Classroom 4015
math.ucdenver.edu/~mkawai/MERC - 303-315-1712

WRITING CENTER

North Classroom 4014 | Campus Village | Auraria Library
writingcenter.ucdenver.edu

UNDERGRADUATE ACADEMIC ADVISING

ucdenver.edu/undergradadvising

*Graduate students: contact your graduate program
directly for advising information*

ARCHITECTURE AND PLANNING (CAP)

CU Building 2000
303-315-1000 - cap@ucdenver.edu

ARTS AND MEDIA (CAM)

Arts Building 177
303-315-7400 - camadvising@ucdenver.edu

BUSINESS SCHOOL

15th and Lawrence Street, 4th floor
303-315-8110 – undergrad.advising@ucdenver.edu

CENTER FOR UNDERGRADUATE EXPLORATION & ADVISING (CUE&A)

Student Commons 1113
303-315-1940 - cuea@ucdenver.edu

EDUCATION AND HUMAN DEVELOPMENT (SEHD)

Lawrence Street Center 701
303-315-6300 - education@ucdenver.edu

ENGINEERING, DESIGN AND COMPUTING (CEDC)

North Classroom 3034
303-315-7170 - engineering@ucdenver.edu

LIBERAL ARTS AND SCIENCES (CLAS)

North Classroom 1030
303-315-7100 – clas_advising@ucdenver.edu

PUBLIC AFFAIRS (SPA)

Lawrence Street Center 525
303-315-2228 - spa@ucdenver.edu

Plan Ahead! Review Important Dates & Deadlines at
Go to ucdenver.edu/registrar and select “Academic Calendars”

Academic Calendar - Summer 2021

Main Session	Date	Important Notes
First day to apply for Summer Graduation via UCDAccess.	March 1, 2021	
Registration begins for Summer Semester via UCDAccess.	March 1, 2021 - March 16, 2021	Check UCDAccess for your specific registration date and time assignment. For best course selection, register as soon as possible after your registration time assignment.
Open enrollment begins.	March 17, 2021	
Registration begins for Fall 2021 Semester via UCDAccess.	April 1 - April 16, 2021	Check UCDAccess for your specific registration date and time assignment. For best course selection, register as soon as possible after your registration time assignment.
Open enrollment begins for Fall 2021 Semester.	April 19, 2021	
First day of Summer semester classes.	June 7, 2021	
Last day to waitlist classes using UCDAccess.	June 10, 2021	
Last day to drop a class without a \$100 drop charge (by 11:59 PM MT).	June 11, 2021	All waitlists will be eliminated today.
Census	June 15, 2021	
Last Day to add classes in UCDAccess.	June 15, 2021	
Last day to drop classes with a financial adjustment (before 5pm).	June 15, 2021	
Classes dropped after 5pm will appear on your transcript with a grade of 'W'.	June 15, 2021	
Full tuition will be charged for additional classes added after 5pm.	June 15, 2021	College Opportunity Fund hours will not be deducted from eligible student's lifetime hours.
Last day to apply for Summer graduation via UCDAccess.	June 15, 2021	After this date, contact your advisor.
Last day to request or cancel Grade Forgiveness.	June 15, 2021	Refer to the Grade Forgiveness form for restrictions.
Independence Day Observed	July 5, 2021	Campus closed.

Last day to withdraw from a class via UCDAccess.	July 11, 2021	
Last day to request No Credit or Pass/Fail grade for a class.	July 11, 2021	Graduate degree students can exercise the P/F option for undergraduate courses only. Graduate students should consult their school or college regarding the P/F option. A grade of P will not be acceptable for graduate credit to satisfy any Graduate School requirement.
First day to withdraw from a class with a Late Withdraw Petition form.	July 12, 2021	
Last day to withdraw from a class with a Late Withdraw Petition form.	July 21, 2021	
End of semester.	July 31, 2021	
Final grades available on UCDAccess and transcripts (tentative).	August 5, 2021	
Summer degrees posted on UCDAccess and transcripts (tentative).	August 27, 2021	This is the date your degree will be recorded on your transcript; diplomas begin mailing on September 15th.

Important Information

Refer to the Residency website for important deadlines pertaining to In-State Tuition Rate qualification.

Refer to the College Opportunity Fund (COF) website for important deadlines pertaining to the COF stipend for eligible undergraduate students paying in-state tuition.

Additional Billing/Financial Information: Students are responsible for complying with tuition/fees deadlines. All registered students must access their student account and billing information through UCDAccess. You will also receive an electronic bill to your university email account.

Intensive and Module classes require the same amount of work and number of classroom hours as full-term classes. Intensive classes are less than five weeks. Module classes last five or more weeks, but less than full term. Module/intensive classes may be added up until the first day of the class. After the first day of class, late starting module or intensive classes may be added with the instructor's signature approval. Instructor approval is not required to drop the class within the first 15% of class meetings.